

ARDMORE, OK, USA

WMO: 723529

Lat: 34.147N

Lon: 97.123W

Elev: 257

StdP: 98.27

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 53929

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.0	-3.9	-14.1	1.1	-1.5	-12.0	1.4	0.4	8.2	9.3	7.5	10.2	1.8	20	0.409

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	38.6	23.7	37.1	23.3	35.2	23.0	25.9	32.2	25.5	32.0	25.1	31.5	3.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.7	28.2	23.9	19.3	28.0	23.0	18.3	27.5	81.8	32.2	79.5	32.3	77.5	31.2	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.8	5.8	-9.9	39.7	3.4	2.4	-12.4	41.4	-14.4	42.8	-16.3	44.1	-18.8	45.9		
(5)			-10.8	26.3	3.1	1.2	-13.0	27.2	-14.8	27.9	-16.6	28.6	-18.8	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	6.5	8.2	13.0	17.3	21.6	26.1	28.6	28.6	24.6	18.2	12.6	7.3
	DBStd	9.03	5.40	5.88	5.24	4.51	4.04	2.62	2.57	2.92	3.39	4.67	5.25	4.96
	HDD10.0	413	135	93	29	4	1	0	0	0	0	3	33	114
	HDD18.3	1507	368	284	176	71	18	0	0	0	2	61	182	344
	CDD10.0	3244	26	44	123	223	359	482	576	577	438	257	110	29
	CDD18.3	1297	1	2	12	39	118	232	318	319	190	56	9	1
	CDH23.3	14553	8	20	100	320	1051	2458	4049	4007	1933	516	81	8
	CDH26.7	7050	1	3	16	68	343	1099	2221	2252	879	156	13	0
Wind	WSAvg	2.4	2.3	2.7	2.9	3.1	2.8	2.5	2.1	1.9	1.9	2.1	2.3	2.2
Precipitation	PrecAvg	906	44	51	72	85	126	102	58	65	106	101	64	56
	PrecMax	1397	156	165	183	344	341	240	172	298	268	428	170	182
	PrecMin	376	0	0	10	7	6	15	0	0	3	0	1	3
	PrecStd	225	38	39	44	66	83	57	48	63	69	83	50	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.1	26.2	28.0	31.3	34.1	37.2	41.2	41.3	37.5	32.5	28.0	24.0
		MCWB	14.6	16.0	17.3	19.3	23.1	24.2	23.3	23.2	21.8	20.6	18.4	16.0
	2%	DB	21.3	22.6	26.2	28.0	32.3	35.0	38.0	39.0	34.9	30.3	26.0	21.1
		MCWB	13.3	14.2	16.8	19.2	22.9	23.4	23.7	23.4	22.1	20.4	17.0	15.0
	5%	DB	18.0	20.2	23.7	27.0	30.1	32.9	37.0	37.4	32.8	27.9	22.9	17.9
		MCWB	11.9	14.0	16.1	18.6	21.7	23.6	23.4	23.5	22.1	19.2	16.2	13.0
	10%	DB	16.2	17.7	22.0	25.1	28.0	32.2	35.1	36.0	32.0	26.3	21.3	16.2
		MCWB	10.0	12.7	15.1	17.4	21.5	23.3	23.2	23.2	21.7	18.6	15.7	11.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.1	18.8	19.9	23.4	25.8	26.8	26.6	26.4	26.0	23.5	20.9	19.5
		MCDB	20.5	21.7	24.9	27.4	30.7	32.1	33.3	32.7	30.3	28.3	24.6	21.5
	2%	WB	16.0	17.1	18.7	21.7	24.6	25.8	25.8	25.7	25.0	22.3	19.5	16.9
		MCDB	19.0	20.0	23.2	25.2	29.2	31.5	32.7	32.8	29.9	27.3	23.5	19.3
	5%	WB	12.9	15.4	17.7	20.5	23.7	25.2	25.4	25.2	24.1	21.4	17.9	14.6
		MCDB	17.3	19.3	21.5	24.2	28.2	30.8	32.4	32.7	28.9	25.8	21.1	17.1
	10%	WB	10.4	13.1	16.6	19.5	22.7	24.5	25.0	24.6	23.3	20.4	16.5	11.7
		MCDB	15.3	17.6	20.5	23.7	27.2	29.6	31.8	32.0	28.4	24.3	19.8	14.9
Mean Daily Temperature Range	5% DB	MDBR	11.8	11.4	11.6	11.8	10.7	10.3	11.4	11.9	11.8	12.2	11.9	11.0
		MCDBR	16.1	15.0	13.9	13.0	12.4	11.5	13.7	14.0	13.5	13.9	14.0	14.7
		MCWBR	9.6	9.1	7.1	6.7	5.2	3.9	3.4	3.4	4.3	6.0	7.3	9.3
	5% WB	MCDBR	13.6	12.8	11.0	10.4	10.3	9.8	11.1	11.6	10.4	10.4	11.3	11.4
		MCWBR	9.8	9.7	6.9	6.9	5.5	4.1	3.7	3.5	4.4	5.5	7.8	9.5
Clear-Sky Solar Irradiance	taub	0.304	0.312	0.350	0.380	0.416	0.429	0.432	0.432	0.391	0.349	0.325	0.306	
	taud	2.500	2.492	2.408	2.344	2.310	2.312	2.340	2.310	2.380	2.485	2.490	2.496	
	Ebn at Noon	911	940	924	907	873	857	853	848	870	885	877	885	
	Edh at Noon	84	94	111	124	130	130	126	127	113	94	85	80	
All-Sky Solar Radiation	RadAvg	2.81	3.36	4.44	5.43	5.83	6.60	6.73	6.15	5.11	3.95	2.99	2.43	
	RadStd	0.29	0.51	0.30	0.43	0.60	0.66	0.40	0.36	0.49	0.57	0.38	0.29	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (45 neighbors)		(m)								
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page